LECTURE 5

Approximate Solution of Algebraic and Transcendental Equations

What is the solution of the equation 3x+5=7?

Solution: 3x=2 then x=2/3.

Remember: x=a is a root of a function if f(a)=0

What is the root of the function f(x)=2x-15?

Solution: f(x)=2x-15=0 then x=15/2.

For years, one of important problems of mathematicians is to find a root of a function (to find the solution of an equation).

There are many functions that finding the roots is not so easy. So that many methods developed for to find or to approximate the roots of functions.

But before trying to find a root, first we should decide a root exist or not. For the existence of a root let us state the important theorem from calculus 1:

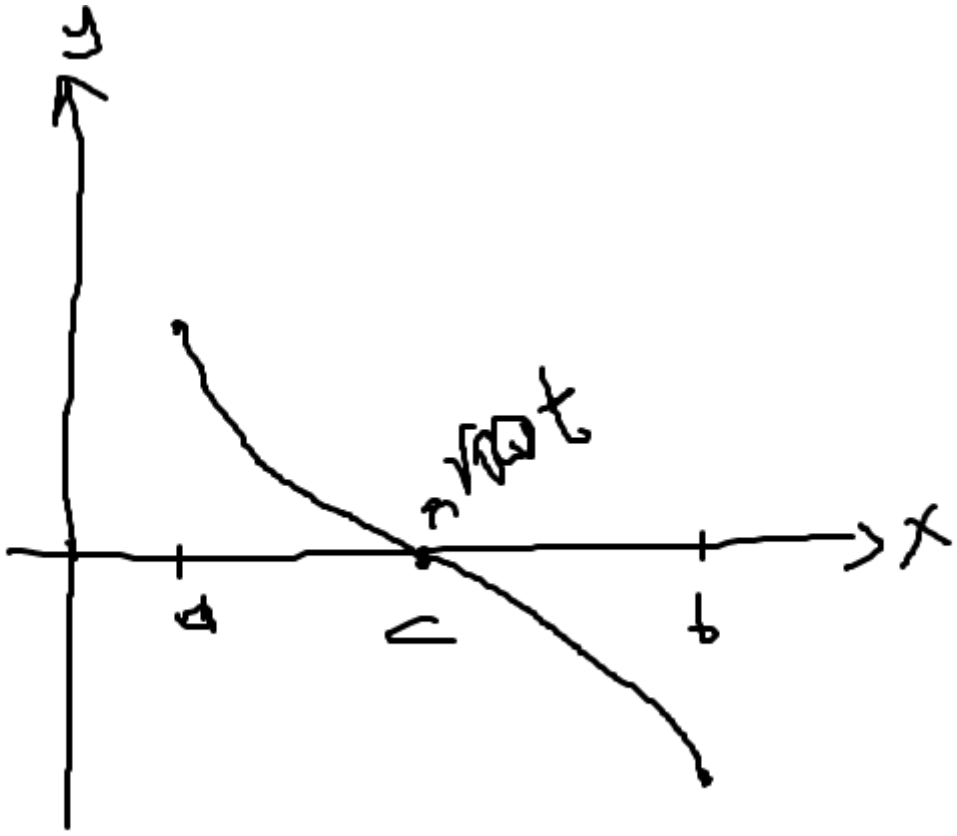
Intermediate Value Theorem: Let f(x) be continuous on a closed interval [a,b].

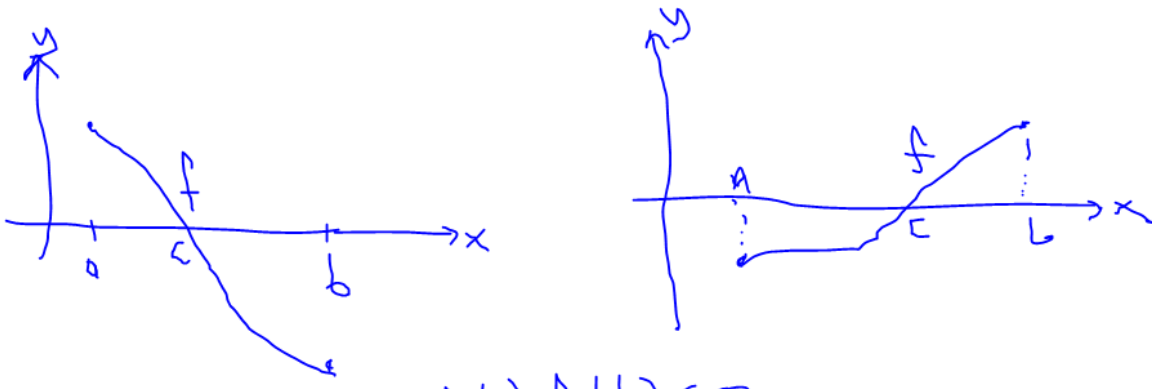
Assume f(a)f(b)<0 (one is positive, one is negative)

Then therefore there exists at least one point $c \in (a, b)$

Such that f(c)=0.

Note: In this way we know that the root c is between a and b.





$f(a) \cdot f(b) < 0$

But we don't know what is c?
But we know it exists.

Now we need to approximate the root of a function.

To do this there are many methods.

The first method is:

Bisection Method:

I will explain the bisection method with an example:

Example: Use the bisection method to find a root of the equation

$x^3-4x-8.95=0$ accurate to three decimal places.

Solution: $f(x)=x^3-4x-8.95$

$f(3)=27-8-8.95>0$ and $f(1)=1-4-8.95<0$

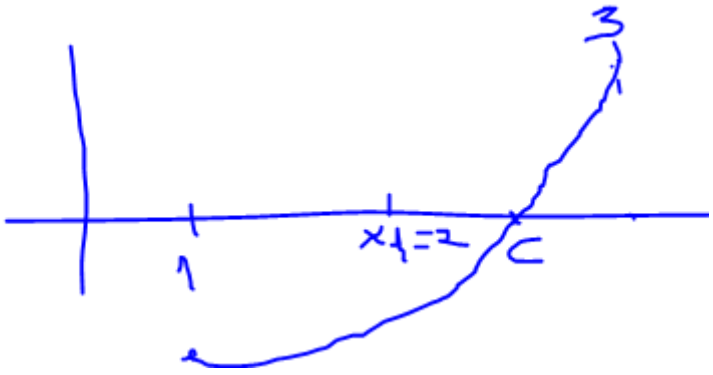
f(x) is continuous (because it is a polynomial) on the interval [1,3] and f(1).f(3)<0 so that , by the intermediate value theorem, there exist at least one root between 1 and 3.

Let us find an approximation for this root using bisection method:

The first approximation for the root is:

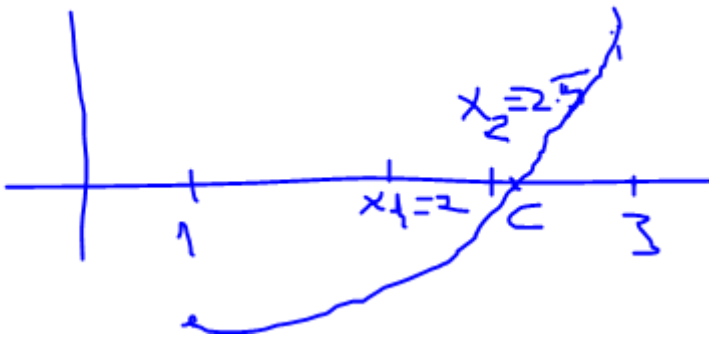
$x_1=(1+3)/2=2,$

$f(2)=8-8-8.95=-8.95<0$



Then the root is in the interval [2,3] since $f(2)f(3)<0$

So the next approximation is $x_2=(2+3)/2=2.5$



$f(2.5)=(2.5)^3-4(2.5)-8.95=(2.5)^3-18.95=15.625-18.95<0$

2 2.5 3

$f(2)<0$ $f(2.5)<0$ $f(3)>0$

So the root is in the interval [2.5, 3] since $f(2.5)f(3)<0$

So our third approximation is $x_3=(2.5+3)/2=2.75$

$f(2.75)=(2.75)^3-4(2.75)-8.95=20.79-11-8.95>0$

2.5 2.75 3

$f(2.5)<0$ $f(2.75)>0$ $f(3)>0$

So the root is in the interval [2.5, 2.75] since $f(2.5)f(2.75)<0$

So our fourth approximation is $x_4=(2.5+2.75)/2=2.625$

$f(2.625)= (2.625)^3-4(2.625)-8.95=18.088-19.4<0$

2.5 2.625 2.75

$f(2.5)<0$ $f(2.625)<0$ $f(2.75)>0$

So the root is in the interval [2.625, 2.75] since $f(2.625)f(2.75)<0$

So the fifth approximation for the root is $x_5=(2.625+2.75)/2=2.6875$

You continue in this way, in each step we get closer closer to the exact value.

When we will stop?

Remember that if we want a approximate value which is accurate at least m significant digits,

We should have $|\varepsilon_a| \leq 0.5 \times 10^{2-m}\%$

Or $|\varepsilon_a| \leq 0.5 \times 10^{-m}$

Let us check at least how many significant digits the approximation x_5=2.6875 is correct.

Let us calculate absolute relative approximate error $|\varepsilon_a|$.

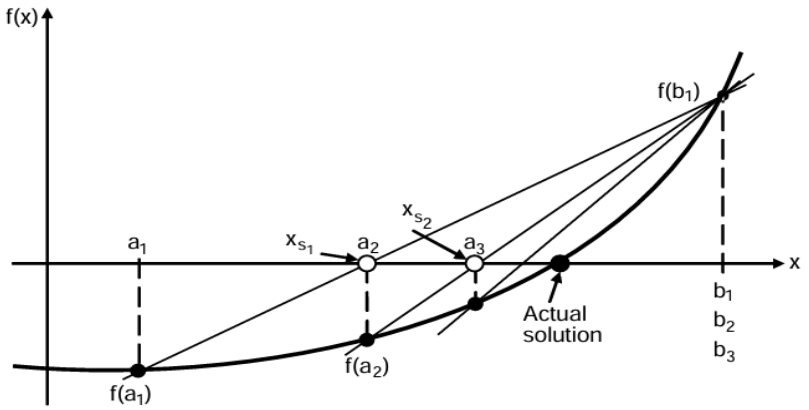
$$|\varepsilon_a| = \left|\frac{present\ app. - previous\ app}{present\ app.}\right|$$

$$|\varepsilon_a| = \left|\frac{2.6875 - 2.625}{2.6875}\right| = 0.023 < 0.05 = 0.5 \times 10^{-1}$$

So we can say right now that our approximation 2.6875 is accurate at least one significant digits using this criteria.

2. THE METHOD OF FALSE POSİTION

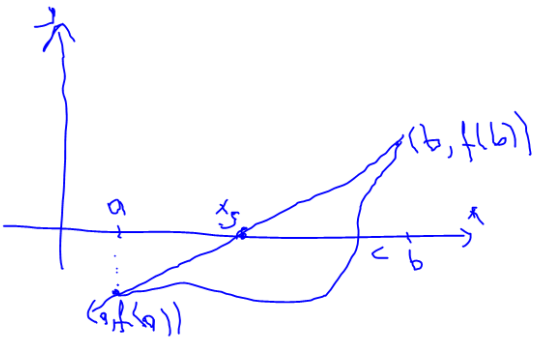
The method of False Position (also called *the Regular Falsi method,* and *the linear interpolation method*) is another well-known bracketing method. It is very similar to Bisection method with the exception that it uses a different strategy to end up with its new root estimate. Rather than bisecting the interval (a, b), it locates the root by joining $f(a_1)$ and $f(b_1)$ with a straight line. The intersection of this line with the x-axis represents an improved estimate of the root.



Algorithm for the method of False Position

1. Define the first interval (a, b) such that solution exists between them. Check $f(a)\, f(b) < 0$.
2. Compute the first estimate of the numerical solution x_s using Eq.(3.8).
3. Find out whether the actual solution is between a and x_{s_1} or between x_{s_1} and b. This is accomplished by checking the sign of the product $f(a)\, f(x_{s_1})$.
   If $f(a)\, f(x_{s_1}) < 0$, the solution is between a and x_{s_1}.
   If $f(a)\, f(x_{s_1}) > 0$, the solution is between x_{s_1} and b.
4. Select the subinterval that contains the solution (a to x_{s_1}, or x_{s_1} to b) is the new interval (a, b) and go back to step 2. Step 2 through 4 are repeated until a specified tolerance or error bound is attained.
   The method of False Position always converges to an answer, provided a root is initially bracketed in the interval (a, b).

In this method the difficulty is to find the x-intercept of the line connecting two points (a, f(a)) and (b,f(b))



Write eqn. of line:

$m = \frac{f(b)-f(a)}{b-a}$

But the point $(x_s, 0)$ is on the line, so $(x_s, 0)$ and $(a, f(a))$ gives the same slope m.

$\frac{0-f(a)}{x_s-a} = \frac{f(b)-f(a)}{b-a} \Rightarrow x_s - a = \frac{b-a}{f(b)-f(a)} \cdot (-f(a))$

$x_s = \frac{-bf(a)+af(a)+af(b)-af(a)}{f(b)-f(a)} = \frac{af(b)-bf(a)}{f(b)-f(a)}$

$$x_s = \frac{a\,f(b) - b\,f(a)}{f(b) - f(a)} \qquad (3.8)$$

Example: Find the root of $f(x)=x^3-4x-8.95$ using false position method. Find the third approximation.

Take a=1 b=3 f(1)=-11.95<0 , f(3)=6.05>0, then f(1)f(3)<0,

Then by the intermediate theorem there exists a roof of f(x) on the interval (1,3). Now let us try to find the apprimate estimate for the root of f(x) using the false position method.

First approximation is: put a=1, b=3

$$x_1 = \frac{1f(3) - 3f(1)}{f(3) - f(1)} = \frac{6.05 + 35.85}{6.05 + 11.95} = \frac{41.90}{18} \sim 2.328$$

1 2.328 3

f(1)<0 f(3)>0

$f(2.328)=2.328^3-4(2.328)-8.95=12.617-9.292-8.95=-5.625<0$

1 2.328 3

f(1)<0 f(2.328)<0 f(3)>0

Hence the root is in the interval [2.328, 3] since f(2.328)f(3)<0

Then by false position method, the second approximation is

Substitute a=2.328 b=3

$$x_2 = \frac{2.328f(3) - 3f(2.328)}{f(3) - f(2.328)} = \frac{2.328(6.05) - 3(-5.625)}{6.05 + 5.625} = 2.65$$

2.328 2.65 3

f(2.328)<0 f(3)>0

$f(2.65)=(2.65)^3-4(2.65)-8.95=18.610-10.6-8.95=-0.94<0$

Hence the root is in the interval [2.65, 3].

Now, let us find the third approximation

$$x_3 = \frac{2.65f(3) - 3f(2.65)}{f(3) - f(2.65)} = \frac{2.65(6.05) - 3(-0.94)}{6.05 + 0.94} = 2.697$$

Now let us check the absolute relative approximate error:

$$|\varepsilon_a| = \left|\frac{present\ app. - previous\ app}{present\ app.}\right|$$

$$|\varepsilon_a| = \left|\frac{2.697 - 2.65}{2.697}\right| = 0.017 < 0.05 = 0.5 \times 10^{-1}$$

In this example, 3 steps, false position method gives better relative approximate error compare to relative approximate error obtained in bisection method in 5 steps. (0.017<0.023). In this example with false position method we approach to the exact value of the root much more quickly.

and false position method

1. Use the Bisection Method to find the root to six correct decimal places. (a) $x^3 = 9$ (b) $3x^3 + x^2 = x + 5$ (c) $\cos^2 x + 6 = x$

2. Use the Bisection Method to find the root to eight correct decimal places. (a) $x^5 + x = 1$ (b) $\sin x = 6x + 5$ (c) $\ln x + x^2 = 3$

and false position method